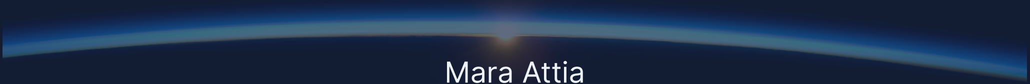
Lecture 5: Atmospheres I
Composition, Structure, & Energy Balance

Planetary Systems



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2026

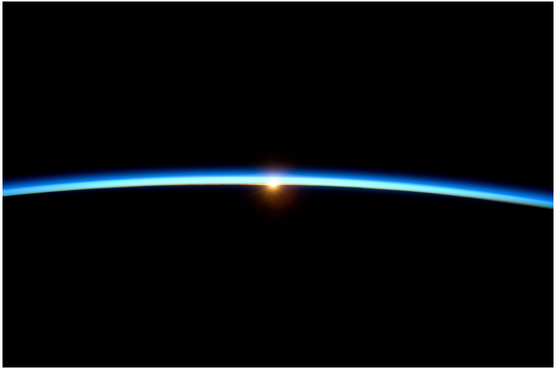
Earth’s thin blue line, ISS STS-129 (2009). Credit: NASA

Learning objectives

By the end of this lecture, you will be able to:

- Classify atmospheres as primary, secondary or tertiary
- Derive the scale height from hydrostatic equilibrium and apply it across the solar system
- Explain the greenhouse effect and the escape mechanisms that decide which atmospheres survive

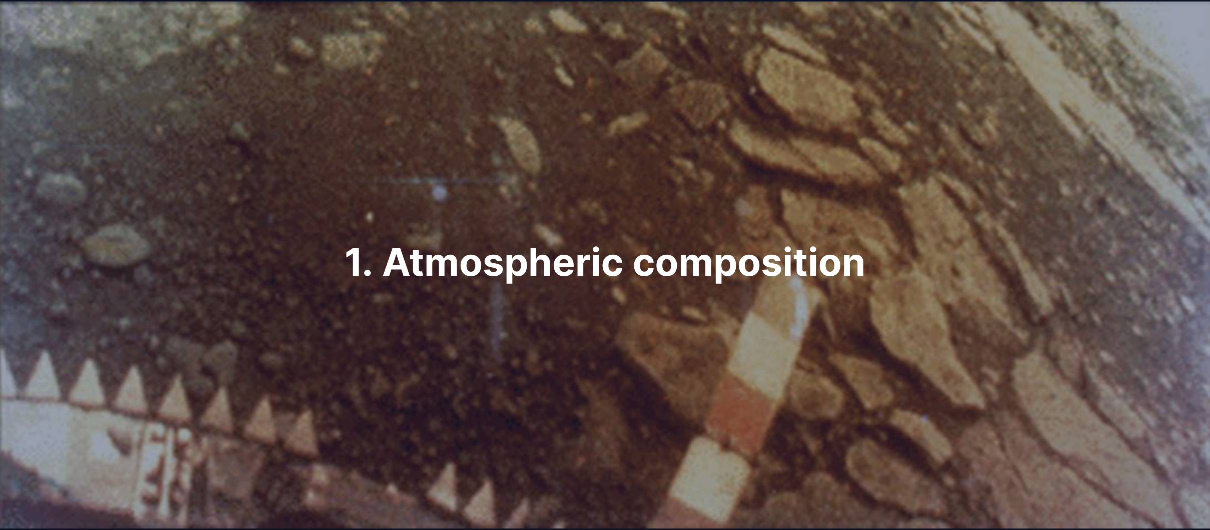
Recap: Lecture 4



Credit: NASA/ISS Expedition 21 crew, public domain

- Metal-silicate separation in magma oceans formed cores within ~30–60 Myr
- A dynamo needs a conducting fluid, convection and $\mathrm{Rm} \gtrsim 10$–100
- Mars lost its dynamo between ~4.1 and 3.7 Ga, then its atmosphere

Today: The thin gaseous envelope that controls a planet's surface conditions.



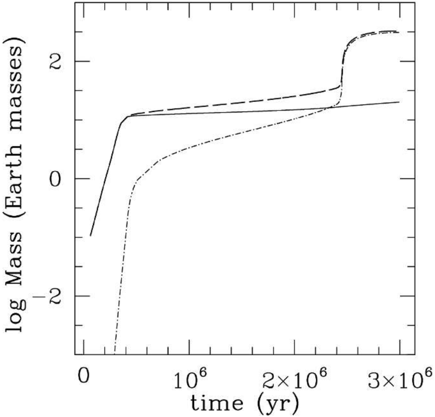
1. Atmospheric composition

The surface of Venus under its 92-bar atmosphere, Venera 13 (1982). Credit: USSR/NASA NSSDC, public domain

Primary atmospheres

Captured directly from the protoplanetary disk.

- Dominated by $\mathrm{H_2}$ and He (solar composition)
- **Gas giants** keep massive $\mathrm{H_2}$/He envelopes; **ice giants** only 10–20% of their mass
- Terrestrial planets are too small to retain primordial $\mathrm{H_2}$/He



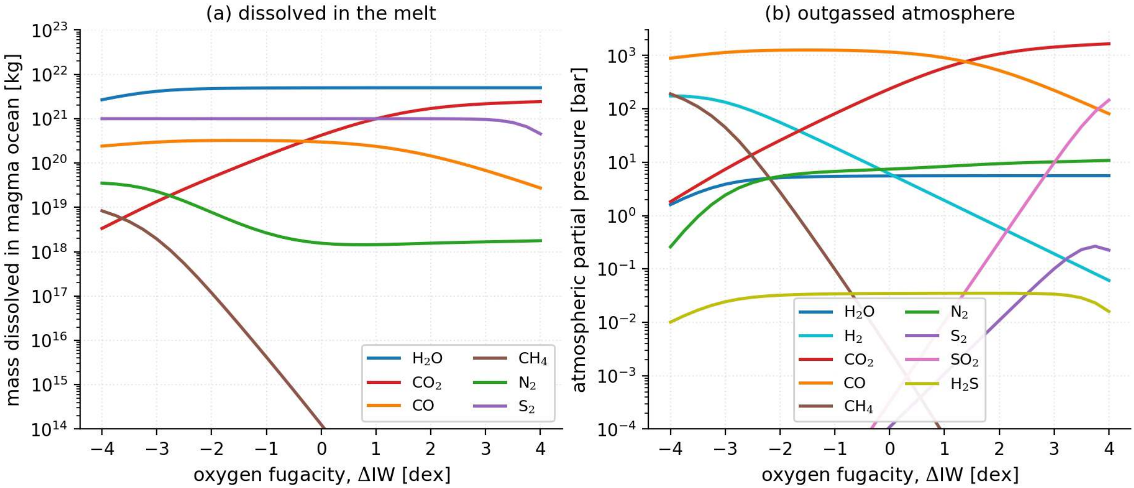
Growth of a giant planet by core accretion: runaway gas accretion builds the H_2/He envelope once it outweighs the core. Reproduced from Helled et al. (2014), Fig. 1; simulation from Lissauer et al. (2009).

A primary atmosphere needs a core of ≳ 5–10 $M_\oplus$ before the disk disperses in ~3–10 Myr.

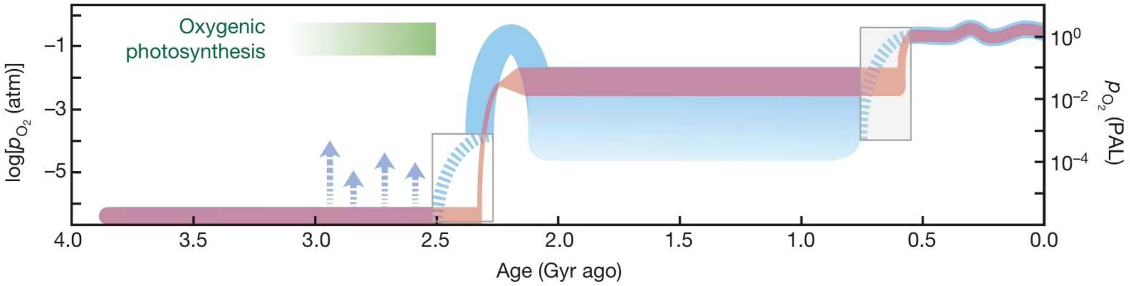
Secondary atmospheres

Produced by outgassing from the interior.

- Magma ocean degassing and volcanism release dissolved volatiles
- Speciation follows the mantle **oxygen fugacity**: oxidising gives $\mathrm{CO_2}$, $\mathrm{H_2O}$, $\mathrm{N_2}$; reducing gives $\mathrm{H_2}$, CO, $\mathrm{N_2}$
- **Venus:** 96.5% $\mathrm{CO_2}$ at 92 bar; **Mars:** 95% $\mathrm{CO_2}$ at 6 mbar; **Titan:** $\mathrm{N_2}$ (from $\mathrm{NH_3}$) at 1.5 bar



Speciation of an outgassed atmosphere against melt oxygen fugacity: CO, H_2 and CH_4 when reducing, CO_2 and H_2O when oxidising. Course figure; model of Nicholls et al. (2024).



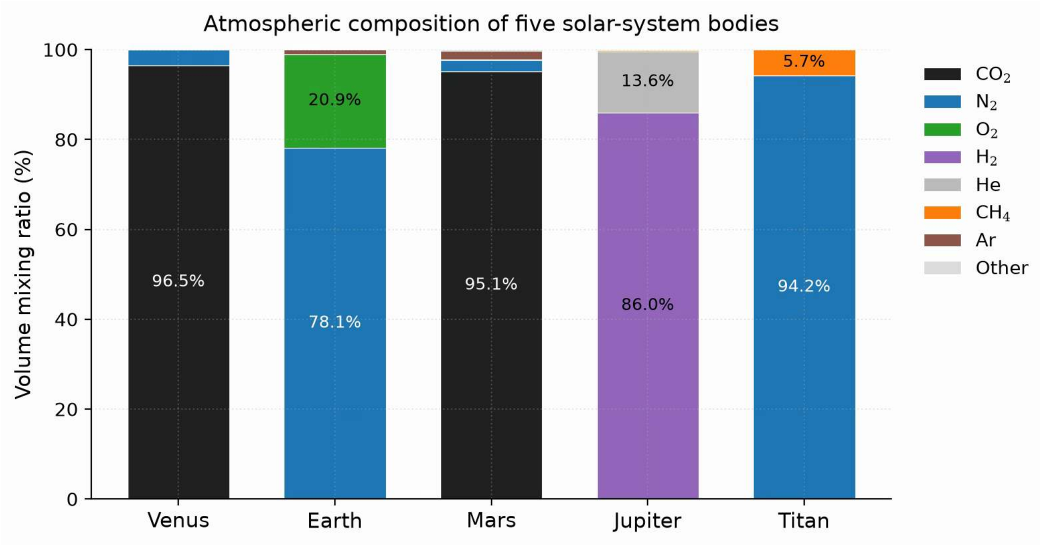
Earth's atmospheric oxygen through time: anoxic for ~2 Gyr, rising at the Great Oxidation Event near 2.4 Ga, modern levels in the Phanerozoic. Reproduced from Lyons et al. (2014).

Earth's atmosphere is **tertiary**: the outgassed $\mathrm{CO_2}$ and $\mathrm{N_2}$ inventory was Venus-like, oxygenic photosynthesis added the 21% $\mathrm{O_2}$, and that $\mathrm{O_2}$ is entirely biogenic and needs active life to persist.

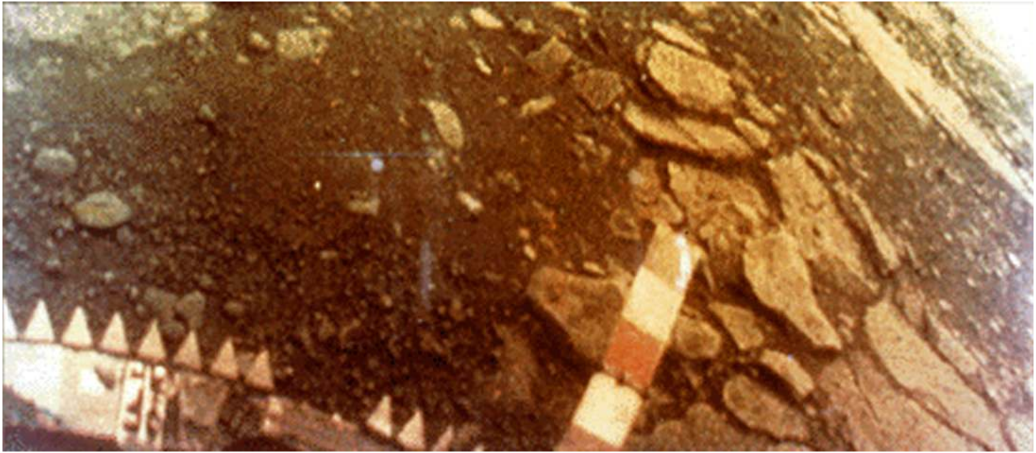
Comparative atmospheric properties

	Venus	Earth	Mars	Jupiter	Titan
P_{surf} (bar)	92	1.0	0.006	n/a	1.5
T_{surf} (K)	737	288	215	n/a	94
Dominant gas	$\mathrm{CO_2}$	$\mathrm{N_2}$	$\mathrm{CO_2}$	$\mathrm{H_2}$	$\mathrm{N_2}$
Secondary gas	$\mathrm{N_2}$	$\mathrm{O_2}$	$\mathrm{N_2}$	He	$\mathrm{CH_4}$
μ	43.4	28.97	43.3	2.2	28.6
Type	2°	3°	2°	1°	2°

Surface pressures span four orders of magnitude, temperatures run from 94 to 737 K.



Atmospheric composition by volume for five solar-system bodies: three origins yield five distinct outcomes. Course figure.



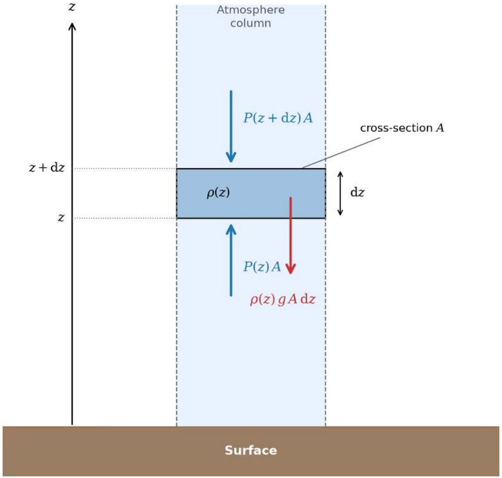
The surface of Venus seen by *Venera 13* on 1 March 1982. Credit: USSR/NASA NSSDC, public domain.

The lander survived 127 minutes at 92 bar and 737 K; the dense CO_2 atmosphere and cloud deck filter sunlight to a dim orange glow.

2. Hydrostatic equilibrium

The thin atmosphere of Mars above its limb, Viking 1 (1976). Credit: NASA/JPL

Hydrostatic equilibrium



Credit: Course figure

The net vertical force on a thin slab of area A and thickness $\mathrm{d}z$ vanishes:

$$P(z)A - P(z+\mathrm{d}z)A = \rho\, g\, A\, \mathrm{d}z$$

$$\boxed{\frac{\mathrm{d}P}{\mathrm{d}z} = -\rho\, g}$$

Each layer supports the weight of all gas above it.

The ideal gas law and barometric formula

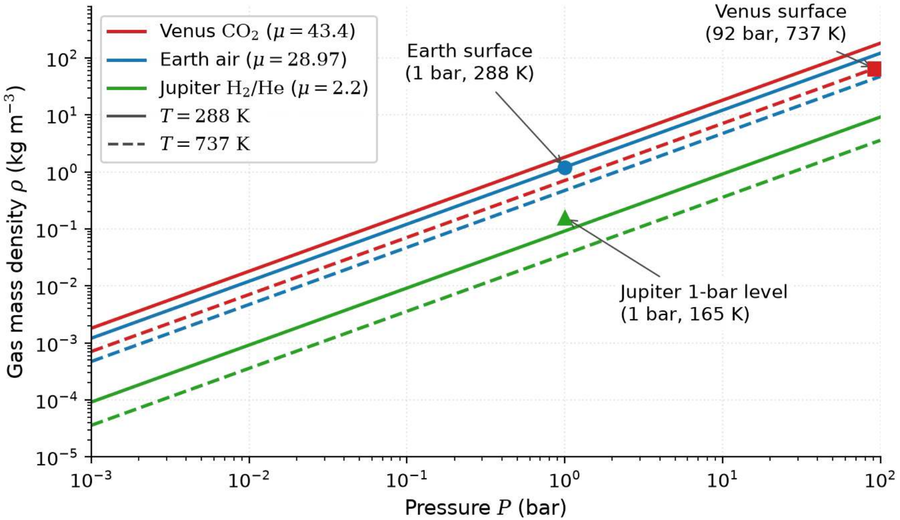
The ideal gas law $P = \rho\, k_{\mathrm{B}} T / (\mu\, m_u)$ in $\frac{\mathrm{d}P}{\mathrm{d}z} = -\rho\, g$ gives

$$\frac{\mathrm{d}P}{\mathrm{d}z} = -\frac{\mu\, m_u\, g}{k_{\mathrm{B}} T} P = -\frac{P}{H}$$

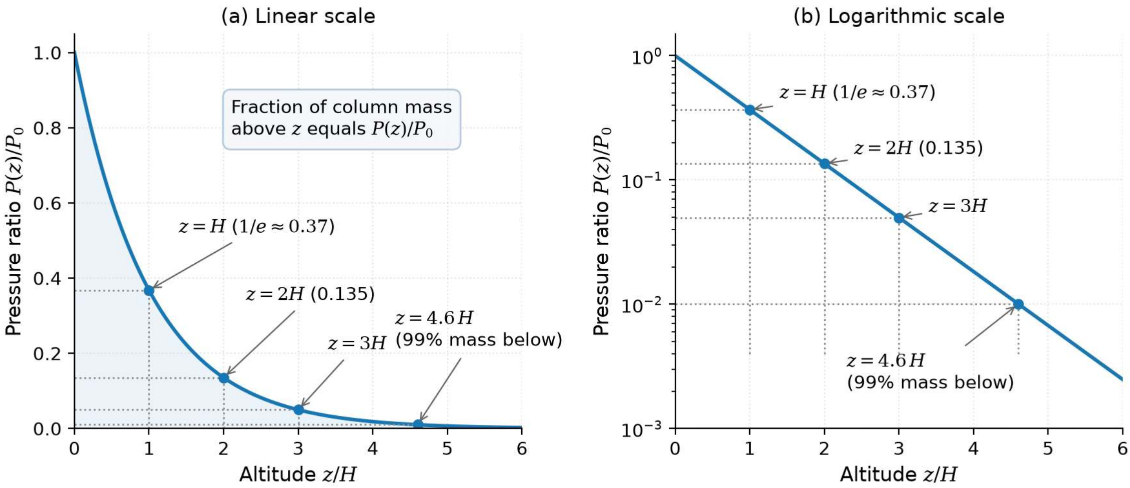
For an **isothermal** atmosphere (constant T):

$$\boxed{P(z) = P_0 \exp\left(-\frac{z}{H}\right)}$$

with $H = k_{\mathrm{B}} T / (\mu\, m_u\, g)$; the pressure drops by a factor of $e \approx 2.718$ each scale height.



Ideal gas law in atmospheric form, $\rho = P\mu m_u/(k_B T)$: density scales with pressure at fixed T, heavier or colder gas is denser. Course figure.

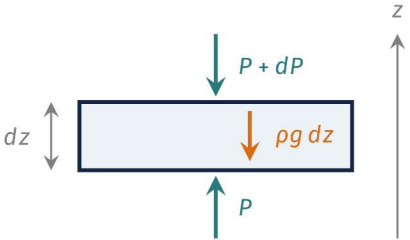


The barometric formula $P/P_0 = e^{-z/H}$: a factor e per scale height, 99% of the column mass below $4.6\,H$, a straight line on a logarithmic axis. Course figure.

3. Blackboard derivation: scale height

Sunrise over Earth’s limb from orbit. Credit: NASA

Goal and strategy



Goal: derive $H = k_{\mathrm{B}}T/(\mu\, m_u\, g)$ from hydrostatic equilibrium and the ideal gas law.

- Balance the forces on a thin slab
- Replace ρ by P with the ideal gas law
- Read off the length scale, integrate, compute

One length, H, sets how fast pressure falls with height.

Deriving the scale height

Start from:

$$\frac{\mathrm{d}P}{\mathrm{d}z} = -\rho\, g \qquad \text{and} \qquad P = \frac{\rho\, k_{\mathrm{B}} T}{\mu\, m_u}$$

Express ρ in terms of P:

$$\rho = \frac{P\,\mu\, m_u}{k_{\mathrm{B}} T}$$

Substitute:

$$\frac{\mathrm{d}P}{\mathrm{d}z} = -\frac{\mu\, m_u\, g}{k_{\mathrm{B}} T} P \equiv -\frac{P}{H}$$

The atmospheric scale height

$$\boxed{H = \frac{k_{\mathrm{B}} T}{\mu\, m_u\, g}}$$

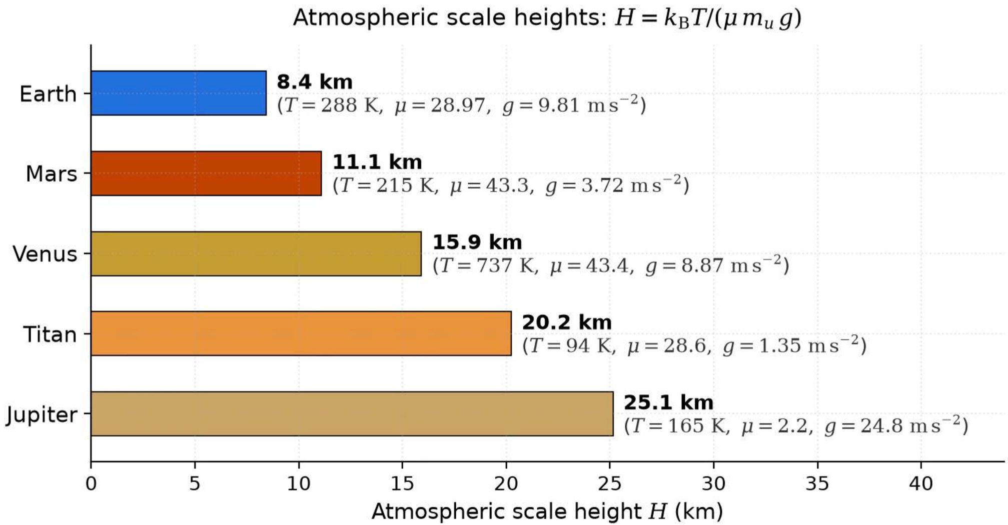
- **Higher** T → larger H: hotter gas extends further
- **Heavier molecules** (larger μ) → smaller H: harder to loft
- **Stronger gravity** g → smaller H: more compressed

H balances thermal energy against gravitational binding; integrated over the column, the surface pressure is the column weight, $P_0 = m_{\mathrm{col}}\, g$.

Scale heights across the solar system

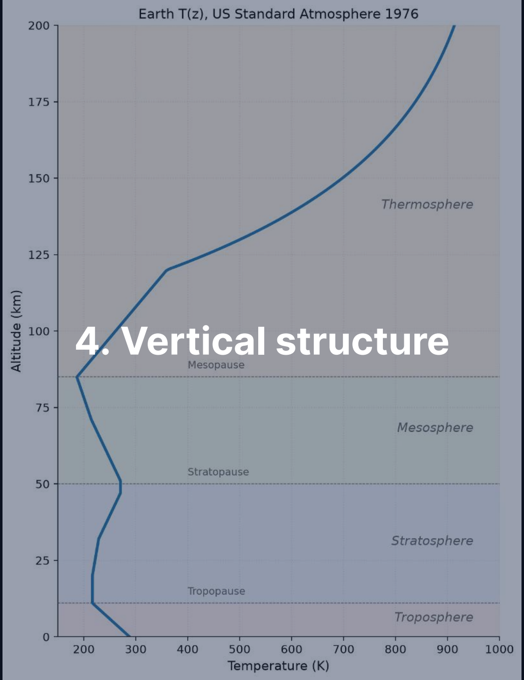
Body	T **(K)**	μ	g **(m s^{-2})**	H **(km)**
Venus	737	43.4	8.87	15.9
Earth	288	28.97	9.81	8.4
Mars	215	43.3	3.72	11.1
Jupiter	165	2.2	24.8	25
Titan	94	28.6	1.35	20

- Jupiter: large H despite strong gravity, because $\mathrm{H_2}$ is very light (μ = 2.2)
- Titan: large H despite low T, thanks to very weak gravity (g = 1.35 m s^{-2})
- Venus: hot, but heavy molecules and decent gravity yield a moderate H

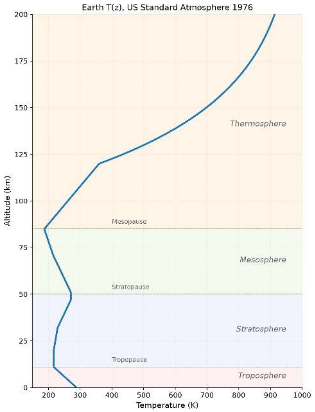


Scale heights recomputed from T, μ and g: light gas (Jupiter) and weak gravity (Titan) give the largest H, Earth the smallest. Course figure.

4. Vertical structure



Earth’s atmospheric layers. Course figure



Earth's atmospheric layers, defined by the sign of $\frac{\mathrm{d}T}{\mathrm{d}z}$. Course figure; US Standard Atmosphere 1976.

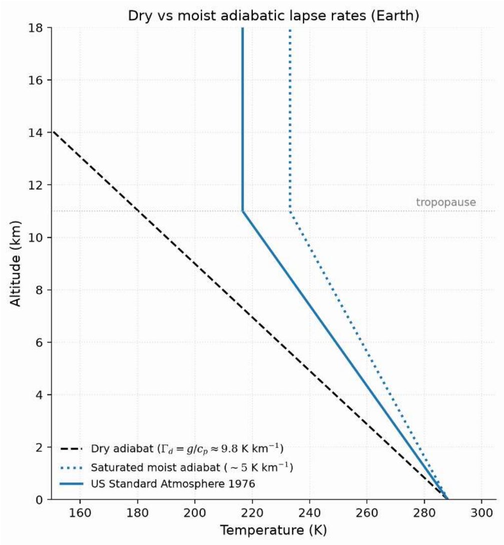
Troposphere (0–12 km): T falls, convective; **stratosphere** (12–50 km): T rises by UV ozone heating; **mesosphere** (50–85 km): T falls to the ~190 K mesopause; **thermosphere** (>85 km): EUV heating, T > 1000 K.

The adiabatic lapse rate

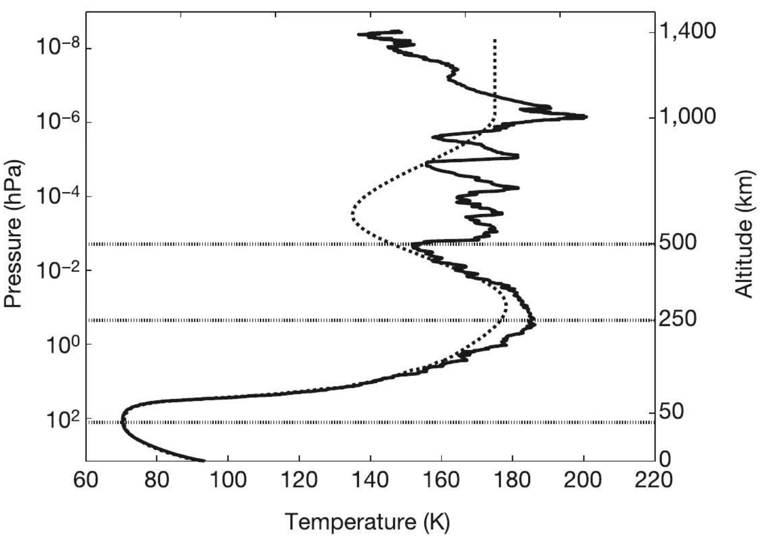
In the troposphere, convection sets the temperature gradient:

$$\Gamma_d = -\frac{\mathrm{d}T}{\mathrm{d}z} = \frac{g}{c_p}$$

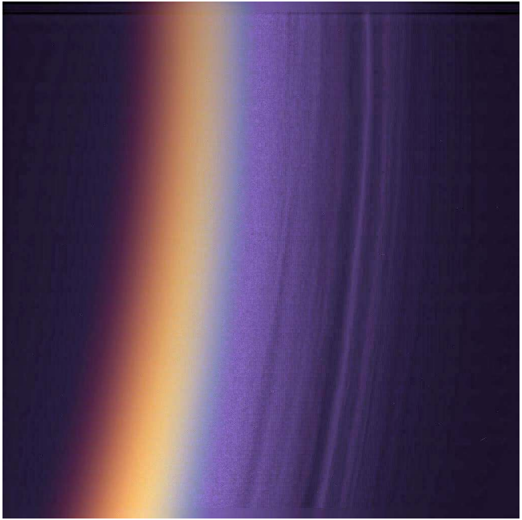
- **Dry adiabatic lapse rate** for Earth: $\Gamma_d = 9.81 / 1004 \approx 9.8\ \mathrm{K\,km^{-1}}$; observed ~$6.5\ \mathrm{K\,km^{-1}}$ because of latent heat
- Rising air expands, does work on its surroundings, and cools
- **Tropopause height:** ~8 km (poles) to ~17 km (equator)



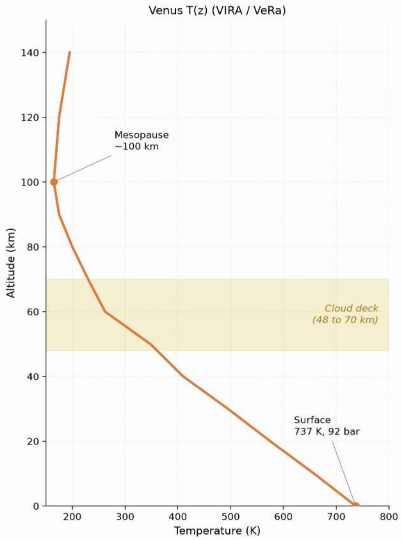
Dry and moist adiabats against the US Standard Atmosphere 1976: latent heat makes the moist adiabat shallower, and real convection sits between both limits. Course figure.



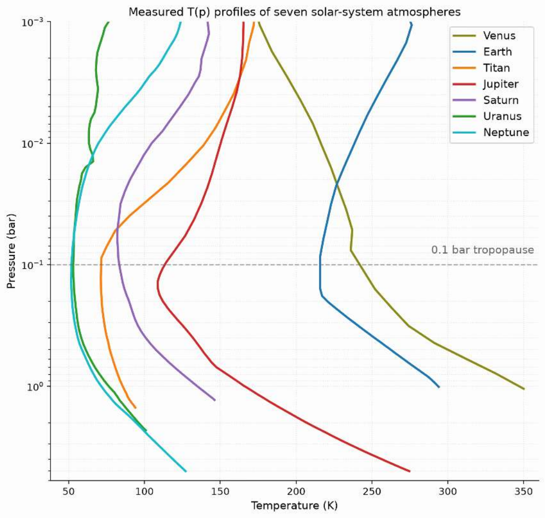
Titan's temperature profile from the Huygens HASI descent: the 94 K surface, the 0.1 bar tropopause and the stratosphere measured in situ. Fulchignoni et al. (2005).



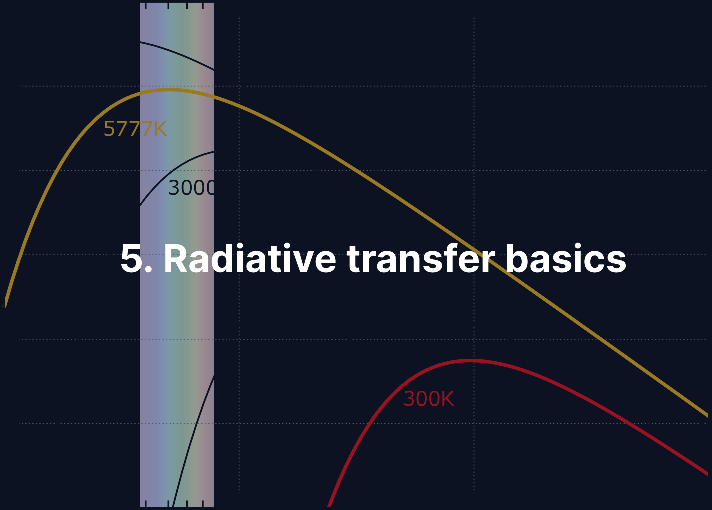
Detached haze layers in Titan’s upper atmosphere, Cassini ISS: photochemical haze up to 500 km altitude absorbs solar UV and heats the stratosphere. Credit: NASA/JPL/Space Science Institute, public domain.



Venus temperature profile from Pioneer Venus/VIRA and Venus Express: without ozone, *T* falls monotonically from 737 K at the surface to the mesopause near 100 km. Course figure.



Measured temperature-pressure profiles of seven solar-system atmospheres: thick atmospheres turn from convective to radiative near 0.1 bar, a universal tropopause. Digitized from Robinson & Catling (2014), Fig. 1.

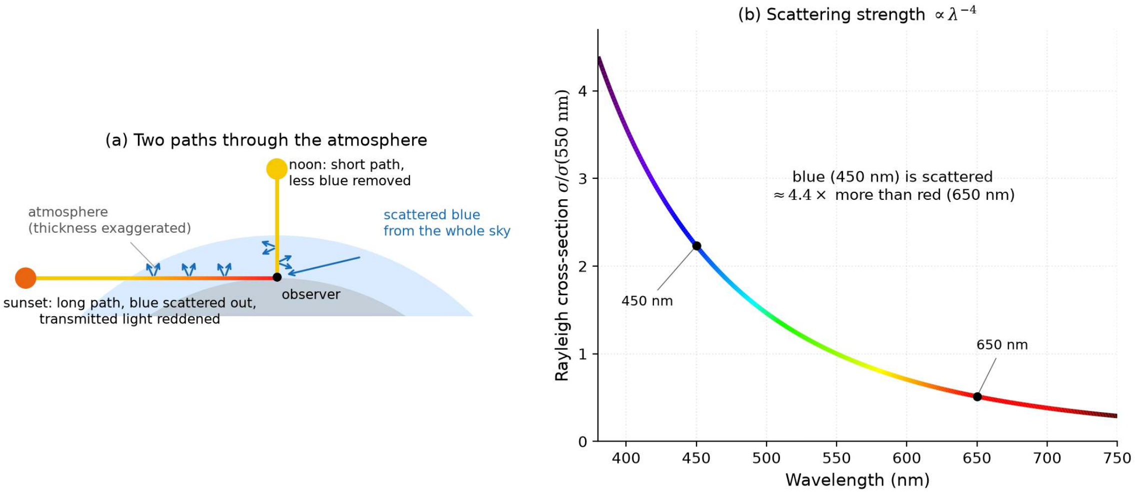


5. Radiative transfer basics

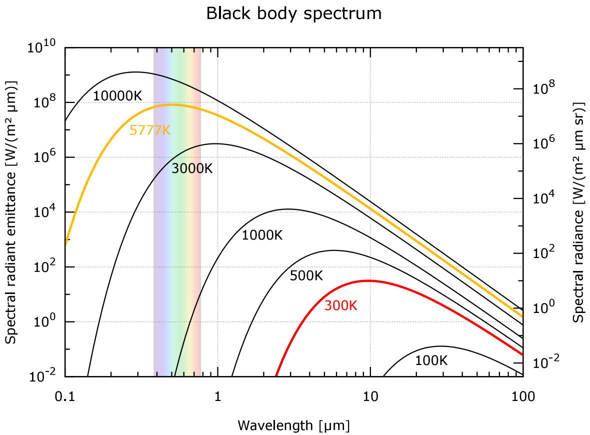
Blackbody spectra. Credit: Wikimedia, public domain

Three interactions: radiation and matter

- **Absorption:** photon energy becomes internal molecular energy; key absorbers $\mathrm{CO_2}$, $\mathrm{H_2O}$, $\mathrm{CH_4}$, $\mathrm{O_3}$, $\mathrm{N_2O}$
- **Emission:** warm gas radiates at the absorbed wavelengths (Kirchhoff's law); only emission from $\tau \lesssim 1$ reaches space
- **Scattering:** photons redirected without absorption; Rayleigh ($\propto \lambda^{-4}$) and Mie scattering



Rayleigh scattering and the colour of the sky: short paths at noon keep the blue scattered light, long slant paths at sunset transmit only red; the cross-section scales as λ^{-4}. Course figure.



Blackbody spectra of the Sun (~5800 K, peak ~0.5 μm) and a planet (~300 K, peak ~10 μm).
Credit: Wikimedia, public domain.

This **wavelength separation** is the basis of the greenhouse effect: an atmosphere can be transparent in the visible but opaque in the infrared.

Optical depth and the Beer-Lambert law

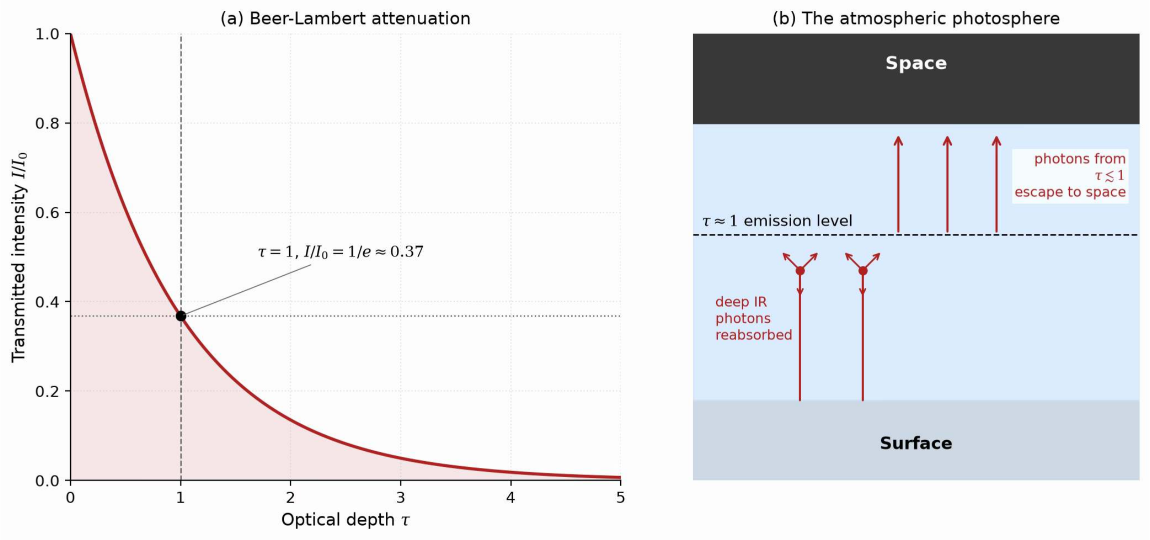
Optical depth τ measures the cumulative absorption along a path:

$$\tau = \int_0^s \kappa \, \rho \, \mathrm{d}s'$$

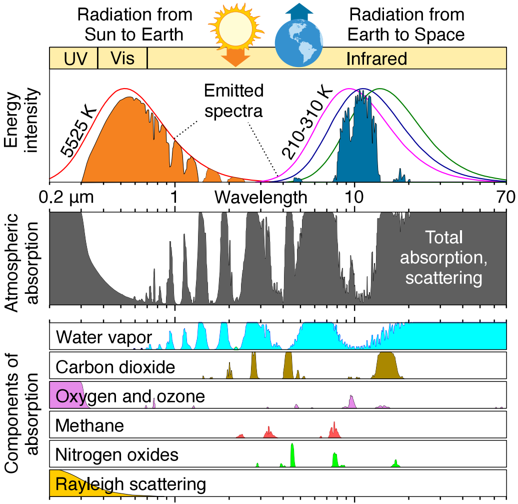
Beam intensity decays following the **Beer-Lambert law**:

$$\boxed{I = I_0 \, e^{-\tau}}$$

- **Optically thin** ($\tau \ll 1$): radiation passes freely through the gas
- **Optically thick** ($\tau \gg 1$): photons are absorbed and re-emitted many times
- **Photosphere** ($\tau \approx 1$): thermal radiation escapes directly to space



The atmospheric photosphere: thermal radiation escapes to space from the $\tau \approx 1$ level, which sets the effective temperature. Course figure.

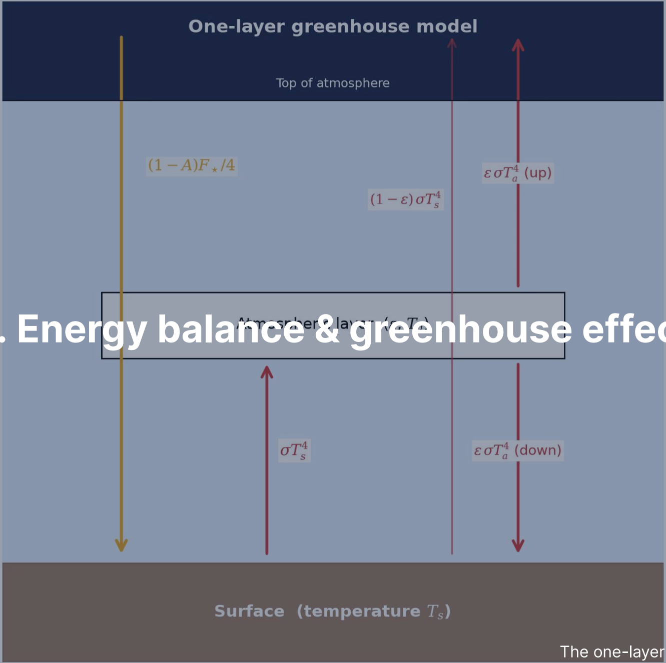


Earth's atmospheric absorption spectrum from UV to IR: greenhouse gases absorb the outgoing infrared while visible sunlight and the 8–13 μm window pass. Credit: Wikimedia Commons, public domain.

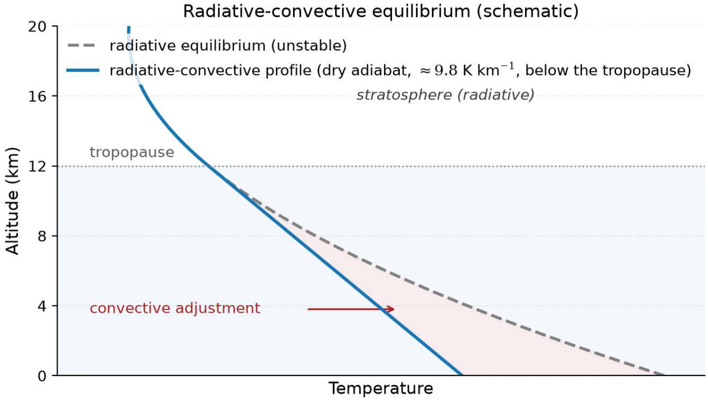
Break

Earthrise, Apollo 8 (1968). Credit: NASA

6. Energy balance & greenhouse effect



The one-layer greenhouse model. Course figure



Radiative-convective equilibrium (schematic): the radiative profile is superadiabatic near the surface, convection holds the troposphere on the dry adiabat. Course figure.

Greenhouse opacity makes the **troposphere** convective, the optically thin **stratosphere** sits in radiative equilibrium, and the **tropopause** is where the radiative gradient becomes shallower than the adiabat.

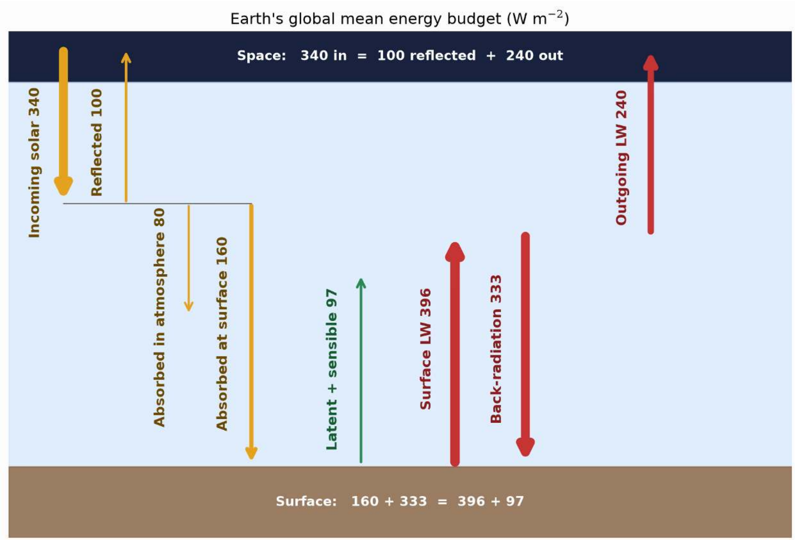
Planetary energy balance

Absorbed stellar power equals emitted thermal radiation:

$$(1 - A)\frac{S}{4} = \sigma T_{\mathrm{eff}}^4 \qquad \text{where} \quad S = \frac{L_\star}{4\pi d^2}$$

- A: Bond albedo; S: stellar flux ($S_0 = 1361\ \mathrm{W\,m^{-2}}$ at Earth)
- Factor $1/4$: intercepting cross-section πR_p^2 over the emitting sphere $4\pi R_p^2$

$$\boxed{T_{\mathrm{eff}} = \left[\frac{(1 - A)\,L_\star}{16\pi\sigma\,d^2}\right]^{1/4} = \left[\frac{(1 - A)\,S}{4\sigma}\right]^{1/4}}$$



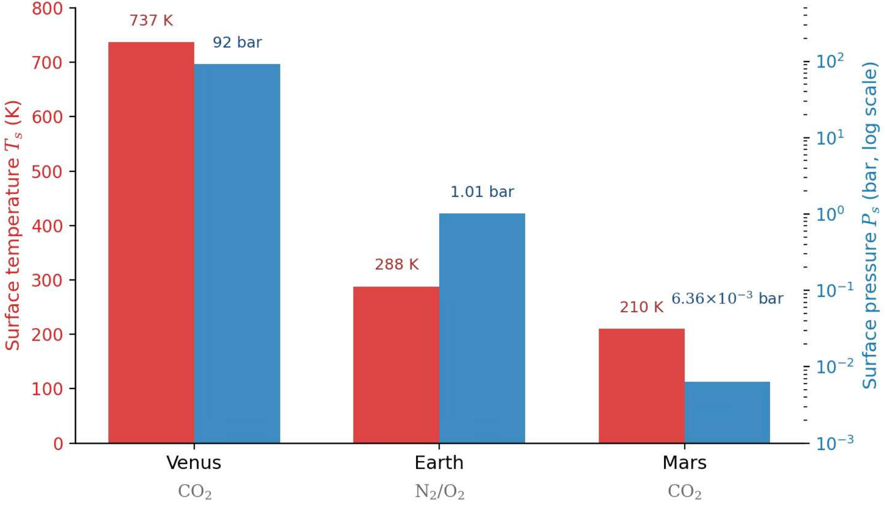
Earth's globally averaged energy budget in W m^{-2}: back-radiation of 333 W m^{-2} supplies more energy to the surface than direct sunlight (160 W m^{-2}). After Trenberth et al. (2009).

Effective vs. surface temperature

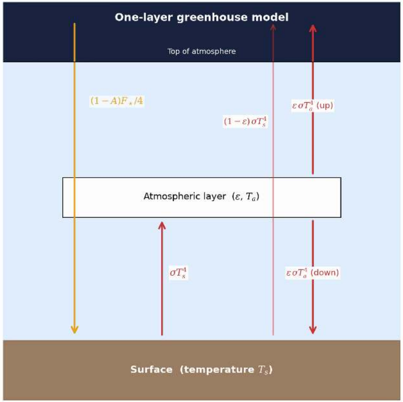
Body	A	d **(AU)**	T_{eff} **(K)**	T_{surf} **(K)**	ΔT **(K)**
Venus	0.77	0.72	227	737	+510
Earth	0.30	1.00	255	288	+33
Mars	0.25	1.52	210	215	+5
Jupiter	0.34	5.20	110	165*	+55

*Jupiter 1-bar level; excess partly from internal heat (Lecture 3).

$\Delta T = T_{\mathrm{surf}} - T_{\mathrm{eff}}$ is the **greenhouse effect**: 510 K on Venus, 33 K on Earth, ~5 K on Mars (6 mbar, no water-vapour amplifier).



Surface temperature and pressure of Venus, Earth and Mars: similar bulk compositions span a factor 10^4 in pressure and 500 K in temperature through greenhouse warming. Course figure; data from NASA.



The one-layer greenhouse model: sunlight in, infrared absorbed and re-emitted half up, half down. Course figure.

Sunlight (~0.5 μm) reaches the surface, the surface emits thermal IR (~10–15 μm), greenhouse gases absorb it and re-emit half downward, and that extra energy makes $T_{\mathrm{surf}} > T_{\mathrm{eff}}$.

One-layer greenhouse model

A single layer with infrared emissivity ε, transparent to visible sunlight:

Flux balance:

Atmosphere: $\varepsilon\sigma T_s^4 = 2\varepsilon\sigma T_a^4$

Surface: $(1 - A)\frac{F_\star}{4} + \varepsilon\sigma T_a^4 = \sigma T_s^4$

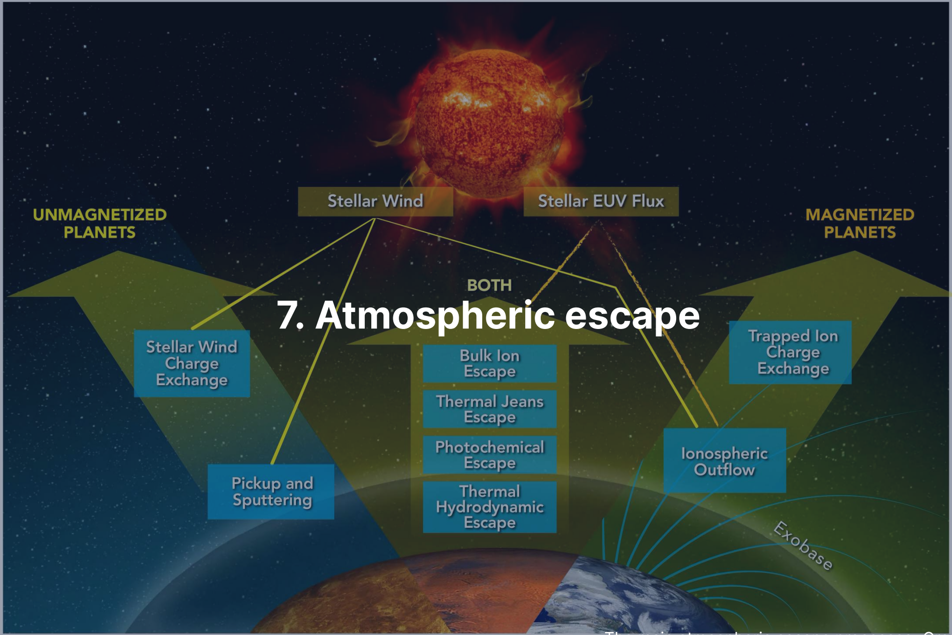
TOA: $(1 - A)\frac{F_\star}{4} = \sigma T_{\mathrm{eff}}^4$

Surface temperature:

$$\boxed{T_s = T_{\mathrm{eff}}\left(\frac{2}{2 - \varepsilon}\right)^{1/4}}$$

- $\varepsilon = 0$ (transparent): $T_s = T_{\mathrm{eff}}$
- $\varepsilon = 1$ (opaque): $T_s = 2^{1/4}\, T_{\mathrm{eff}} \approx 1.19\, T_{\mathrm{eff}}$ (Earth: ≈303 K)

7. Atmospheric escape



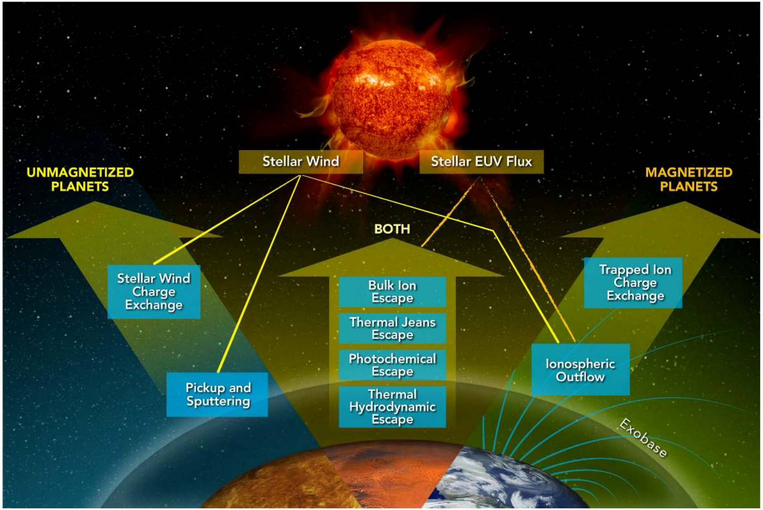
The main atmospheric escape processes. Gronoff et al. (2020)

Thermal (Jeans) escape

The high-velocity tail of the Maxwell-Boltzmann distribution can exceed the escape speed. The **Jeans escape parameter** at the exobase compares gravitational to thermal energy:

$$\lambda_J = \frac{GM_p\, m}{k_{\mathrm{B}} T_{\mathrm{exo}}\, r_{\mathrm{exo}}} = \frac{v_{\mathrm{esc}}^2}{v_{\mathrm{th}}^2}, \qquad v_{\mathrm{th}} = \sqrt{2 k_{\mathrm{B}} T_{\mathrm{exo}} / m}$$

- $\lambda_J \gg 1$: escape is negligible because very few molecules exceed v_{esc}
- $\lambda_J \lesssim 2$–3: thermal escape drives rapid atmospheric loss
- Escape flux $\Phi_J \propto (1 + \lambda_J)\, e^{-\lambda_J}$: exponential in λ_J



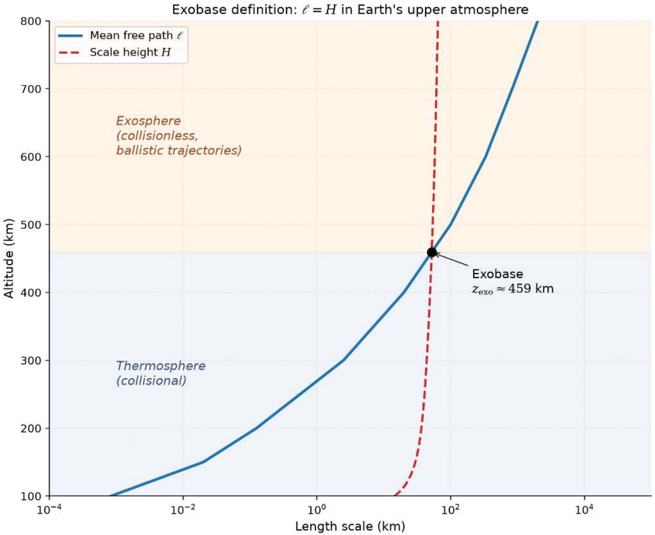
The main atmospheric escape processes: thermal, photochemical and ion channels act on all planets, and the magnetic field selects which dominate. Gronoff et al. (2020).

Jeans parameter for Earth and Mars

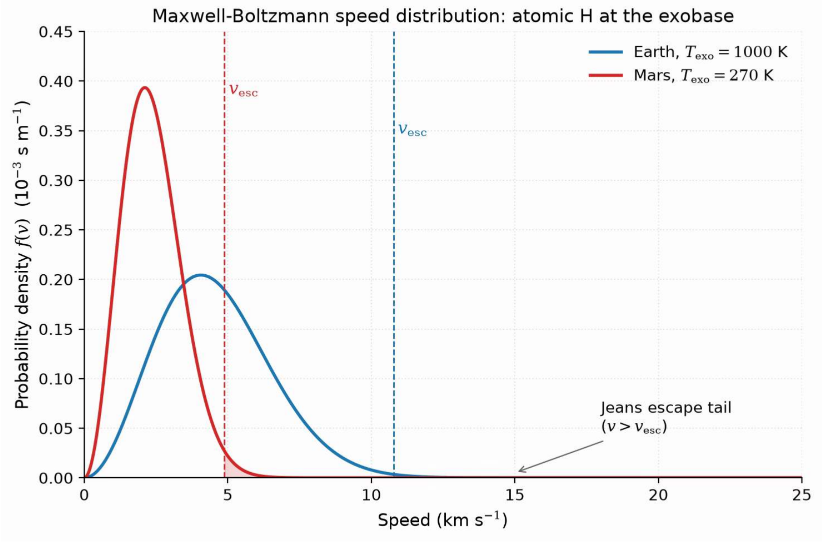
Exobase temperatures: 1000 K (Earth), 270 K (Mars).

Species	m **(u)**	λ_J **(Earth)**	λ_J **(Mars)**
H	1	7.0	5.3
H_2	2	14	11
He	4	28	21
N_2	28	200	150
CO_2	44	310	230

$\lambda_J \propto m$: atomic H escapes steadily from both planets, while N_2 and CO_2 stay gravitationally bound.



The exobase: the altitude where the mean free path ℓ equals the scale height H, collisional fluid below and ballistic flight above. Course figure.



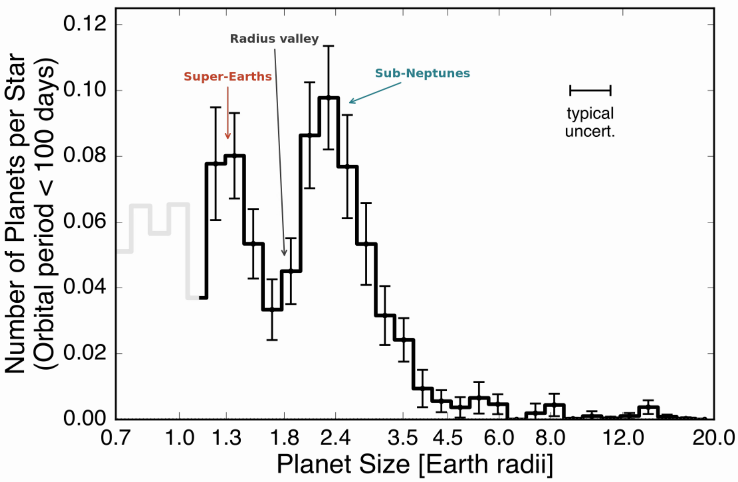
Maxwell-Boltzmann speed distributions for atomic H at the Earth and Mars exobases: the escaping fraction is the tail beyond $v_{\rm esc}$, scaling as $e^{-\lambda_J}$. Course figure.

Hydrodynamic escape

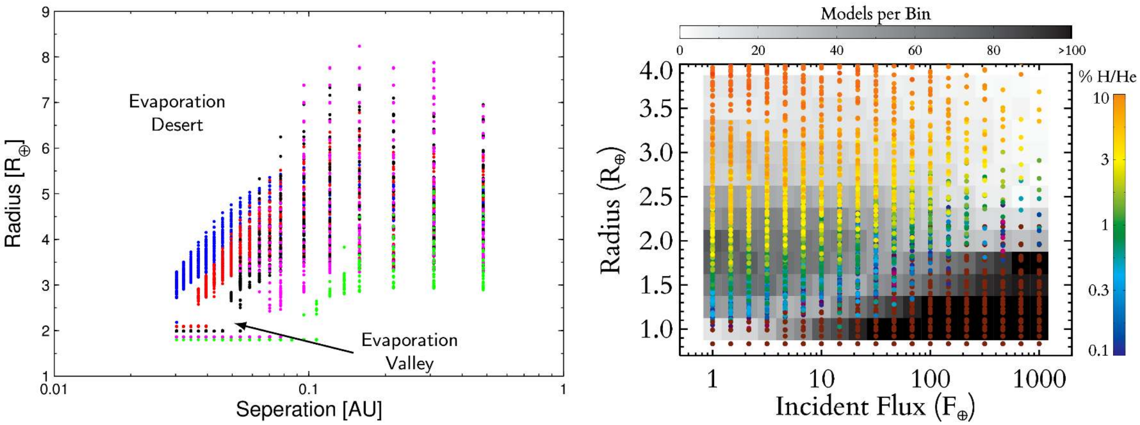
Intense stellar EUV heating drives bulk outflow as an **atmospheric wind**:

- Strongest in the first few hundred Myr, when stellar EUV is 10–100× higher
- Strips $\mathrm{H_2}$-rich primary envelopes from planets up to several $M_\oplus$; outflowing H drags heavier volatiles along
- The primary mechanism behind the exoplanet radius valley at ~1.5–2 $R_\oplus$

Source: Hunten et al. (1987); Owen (2019)



The radius valley in the California-Kepler Survey: a deficit near 1.8 $R_\oplus$ separates stripped rocky cores from envelope-retaining sub-Neptunes. After Fulton et al. (2017).



Population synthesis after 10 Gyr of EUV-driven escape: an empty evaporation desert and a valley near 1.5 to 2 $R_\oplus$. After Owen & Wu (2013), Lopez & Fortney (2013) and Owen (2019).

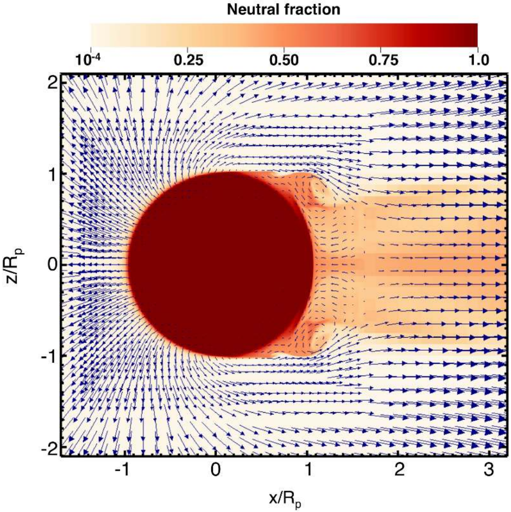
Energy-limited escape rate

Stellar EUV absorbed in the upper atmosphere drives hydrodynamic mass loss:

$$\dot{M}_{\mathrm{EL}} = \eta \frac{\pi F_{\mathrm{EUV}} R_p^3}{G M_p}$$

- Efficiency $\eta \approx 0.1$–0.2 converts incident EUV flux into expansion work
- Scales as R_p^3: extended, low-density atmospheres lose mass fastest
- Scales inversely with M_p: low gravity enables stronger bulk outflow

Source: Watson et al. (1981); Owen (2019)

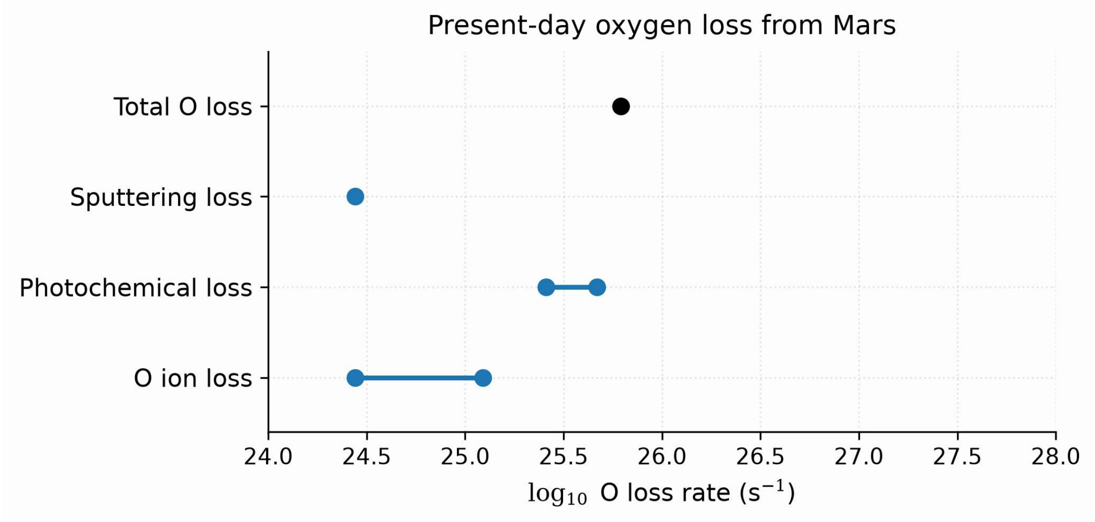


Hydrodynamic outflow from a hot Jupiter under stellar EUV heating: the gas accelerates outward into a night-side wake with Kelvin-Helmholtz instabilities. Reproduced from Owen (2019), Fig. 2; simulation from Tripathi et al. (2015).

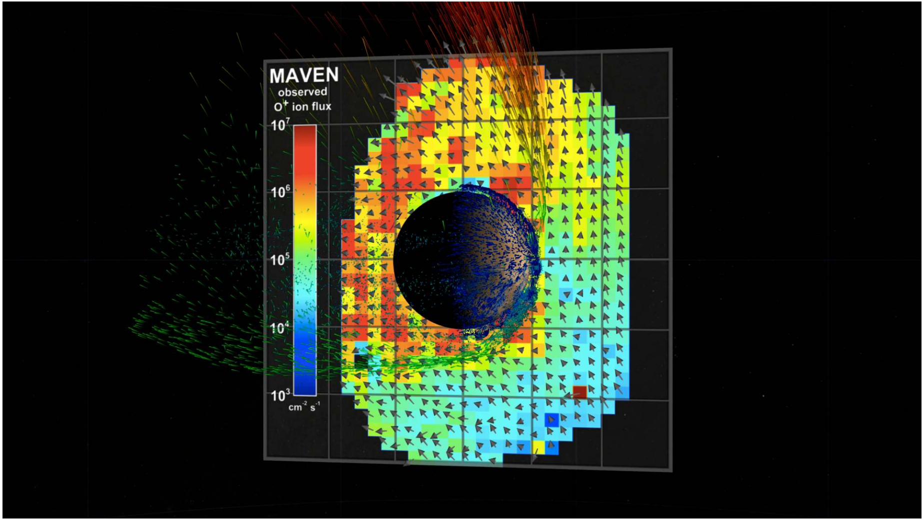
Non-thermal escape mechanisms

- **Sputtering** and **ion pickup:** solar wind ions eject neutrals, and photoions are swept away by the interplanetary field
- **Photochemical:** photodissociation yields suprathermal atoms above v_{esc}
- **Impact erosion:** giant impacts shock and eject atmospheric columns

MAVEN at Mars: photochemical loss of oxygen dominates modern escape over sputtering and ion pickup.

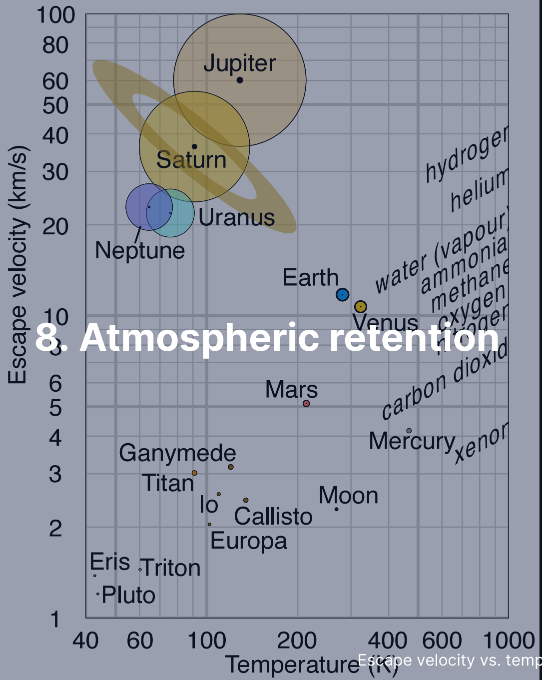


Present-day oxygen loss from Mars by channel: photochemical escape dominates, and with hydrogen escape the total reaches the 2–3 kg s^{-1} MAVEN measures. After Jakosky et al. (2018).

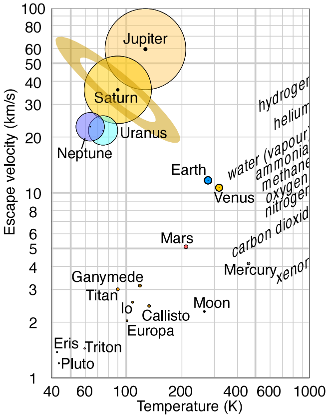


The ion plume of Mars from MAVEN: energetic ions (red) leave in a plume, the bulk (green) along the tail, with no global field to keep the solar wind out. Credit: NASA Scientific Visualization Studio (SVS 4370), public domain.

8. Atmospheric retention

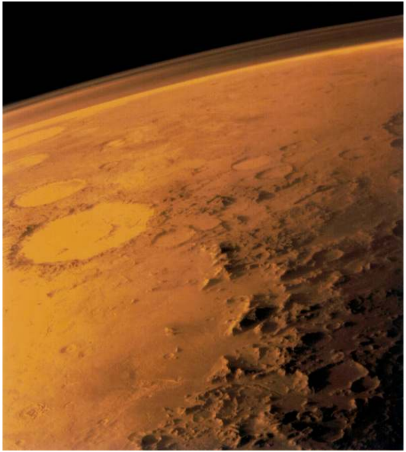


Escape velocity vs. temperature. Credit: Wikimedia, CC BY-SA 4.0



Escape velocity against temperature, with the retention lines for each gas. Credit: Wikimedia, CC BY-SA 4.0.

Long-term retention needs $v_{\rm esc} \gtrsim 6\, v_{\rm th}$: gas giants keep every species, Earth and Venus lose the light gases H and He, Mars is marginal, cold Titan keeps $\mathrm{N_2}$, the Moon and Mercury keep nothing.

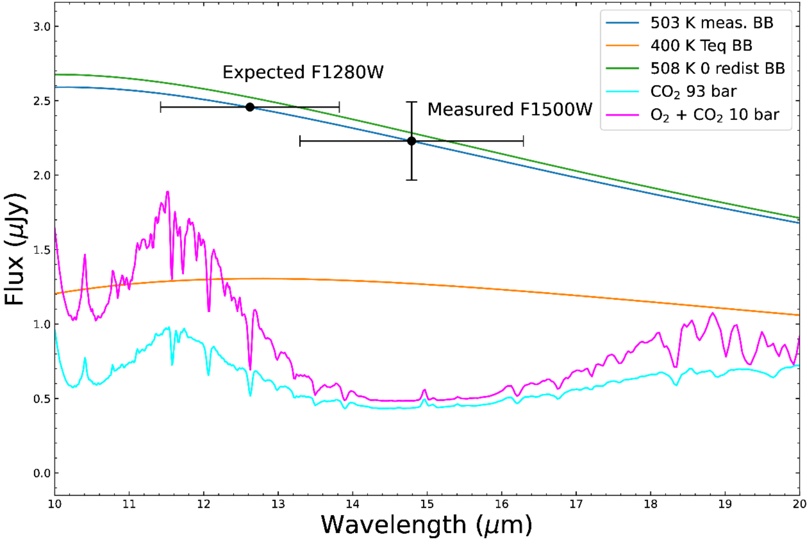


The thin atmosphere of Mars above its limb, Viking 1 (1976). Credit: NASA/JPL.

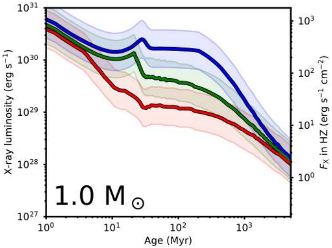
After the dynamo died near ~4.1 Ga the solar wind eroded a dense early atmosphere, recorded by valley networks and lakes, down to today's 6 mbar: no stable liquid water, little greenhouse warming.

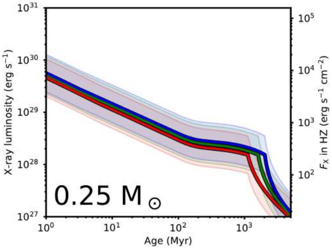
9. Recent advances & outlook

Earth’s atmosphere. Credit: NASA/ISS



JWST/MIRI 15 μm thermal emission matches a 503 K bare rock, ruling out a thick atmosphere on TRAPPIST-1 b. Greene et al. (2023).





X-ray luminosity evolution of Sun-like stars and M dwarfs: low-mass stars stay saturated for over 1 Gyr and strip close-in rocky planets for longer. Reproduced from Johnstone et al. (2021), Fig. 11.

Summary

1. **Atmosphere types**: primary (captured), secondary (outgassed), tertiary (biogenic); four orders of magnitude in surface pressure
2. **Hydrostatic balance** and the **scale height** $H = k_\mathrm{B}T/(\mu m_u g)$ give $P = P_0 e^{-z/H}$; convection sets $\Gamma_d = g/c_p$ below the 0.1 bar tropopause
3. **Greenhouse warming** from $\tau \gtrsim 1$ raises T_s above T_eff; **thermal escape** is negligible when $v_\mathrm{esc} \gtrsim 6v_\mathrm{th}$; hydrodynamic and non-thermal loss follow other rules

Next: Lecture 6

Questions?

Next: Lecture 6, Atmospheres II: clouds, weather & climate (Mara Attia)

Thu 24 Sep, 11:00–13:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 25 Sep, 13:00–15:00, room 5419.0161 (Collegezaal)

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